The Art of Queueing

Reinforcement Learning for visitor flow at the Gallerie degli Uffizi in Florence

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Abstract

The Uffizi Gallery has a congestion problem: demand piles onto a handful of rooms around Botticelli, Leonardo, Raphael-Michelangelo and Caravaggio while the rest of the museum sits empty. We frame it as a reinforcement-learning problem and model it in two layers. A minute-step agent-based simulator moves a mixed crowd through the room graph and scores institutional interventions under a revenue constraint. On top of those crowds, reinforcement-learning agents optimize the individual visit, choosing when to stay, when to move and how far ahead to reserve masterpiece windows under RAMA (Reservation-Anchored Masterpiece Access). Most of the work went into the formulation: appreciation progress in the state, dense closing-time pressure, a crowd-discounted reward and action masking. The full bundle of interventions raises completed welfare by 31% at 5,000 visitors with revenue held. MaskablePPO booking beats every matched baseline. The formulation mattered more than the choice of algorithm.

1 Introduction

Overtourism strains Europe's cultural cities. Venice meters day-trippers, Barcelona caps cruise arrivals and Florence funnels millions of visitors a year into a historic core built for far fewer. At the Uffizi the strain concentrates into a few rooms: on a busy day roughly 5,000 visitors press toward the same handful of galleries, the ones holding the most famous paintings in the world, while most of the museum stays quiet. Relieving that pressure is hard, because the obvious fixes either turn visitors away or cut the revenue the museum depends on.

The economics are those of a common-pool resource. The Uffizi has the floor area to serve many visitors, but utility collapses when demand arrives in the same rooms at the same time. The sharpest bottleneck is the Botticelli chain, where rooms A11 and A12 sit on a forced path and hold *Primavera* and *The Birth of Venus*. Leonardo and Raphael-Michelangelo draw secondary peaks, while whole sections downstairs stay underused. Earlier studies of Uffizi flow [1, 2] optimize these crowds at the institutional level. We ask what a learning agent can add.

We pose two coupled questions. At the museum level, which changes to opening hours, prices and access cut congestion without sacrificing revenue? At the visitor level, how should one pace the visit and how far ahead should one reserve the masterpieces to beat a fixed itinerary? We treat the first as an institutional design problem judged in simulation and the second as a reinforcement-learning problem, linked by a shared model of the crowd.

Our model has two layers. A minute-step agent-based simulator moves a mixed population of 5,000 visitors through a graph of the museum and scores candidate interventions against a revenue constraint. On the crowds it produces, reinforcement-learning agents optimize the individual visit, built on three rungs: a twelve-room tabular toy, the full status-quo walk and a booking task we call RAMA (Reservation-Anchored Masterpiece Access), where the masterpieces sit behind timed windows and the visitor chooses a reservation lead time before entering. Most of the effort went into the formulation rather than the algorithm, putting appreciation progress into the state, adding dense closing-time pressure, discounting crowds inside the reward and masking illegal actions.

Two results stand out. The chosen bundle of eleven interventions raises completed visitor welfare by 31% at the 5,000-visitor crowd while holding ticket revenue, cooling the worst density bands without emptying the museum. Under that redesign a MaskablePPO booking agent beats its matched no-booking baseline in every profile and crowd cell. It learns on its own to reserve earlier as the museum fills. The single most reliable lever was action masking, which mattered more than any tuning once the environment became hard.

Our contribution is threefold. First, we formulate the Uffizi visit as linked population and individual RL problems. Second, we build the RAMA booking MDP with reservation lead time as an action. Third, we document the formulation changes that made learning work, with the failures that forced them. The paper then covers the literature, the virtual museum and its interventions, the three MDPs and their rewards, the experiments and the limitations, with algorithms, hyperparameters and evaluation in the appendices.

2 Related work

Crowd management at the Uffizi has been studied with sensor data and flow models. Attanasio et al. [1] analyze visitor flows in the gallery. Centorrino et al. [2] measure, model and optimize movement in crowded museums. Both optimize flow at the institutional level, whereas we also solve the individual visitor's sequential decision problem. Classic work on how visitors move through an exhibition [6] motivates our heterogeneous profiles and dwell-time behavior. On the method side we use the tools from the course [5]: tabular Q-learning and Double-Q for the toy, deep value-based control with DQN [3] and policy-gradient learning with PPO [4] for the full museum.

3 The Virtual Uffizi

3.1 Graph and calibration

The physical museum is encoded as a directed graph $G = (V, E)$ of 99 nodes: 93 gallery rooms from the floor plan plus six operational nodes, the entry, the three staircases, the panoramic terrace and the exit. Each room carries four central attributes: cultural importance, dwell-time magnetism, legal or effective capacity and section. Ordinary doors are bidirectional, while staircases and exit routes are directed. The topology is validated with structural invariants: weak connectivity, reachability from the entry, reachability of an exit from every gallery, high degree for the main corridor hubs and a forced Botticelli chain A10–A11–A12–A13.

Visitors arrive through 37 fifteen-minute entry slots with a broad morning-weighted envelope. The simulator uses a one-minute time step, a 615-minute baseline day (08:15–18:30) and a final-entry cutoff at 17:30. The full intervention scenario extends the day to 720 minutes. The population mix is 60% Instagram tourists, 30% normal tourists and 10% art lovers. Instagram tourists concentrate on famous rooms and the panoramic terrace, move quickly and are nearly crowd-indifferent. Normal tourists visit the famous rooms plus a few canonical additions such as the Tribune and Caravaggio. Art lovers have heterogeneous taste vectors, longer dwell times, stronger crowd aversion and a higher probability of accepting alternative routes.

This mix was itself a correction. An early version of the simulator used a single guidebook-tourist profile. That choice averaged away the pressure we care about: with everyone behaving identically, the Botticelli, Leonardo and Raphael peaks flattened into a mild general load. Splitting the population into the 60/30/10 mix made the crowd process part of the environment and restored the bottleneck the project is about.

The movement model is local and stochastic. Conditional on deciding to move, a visitor chooses among neighbors with weights of the form

$$w_{i\to j} = \max\{0.01,\, b_{\text{route}}(j) + b_{\text{importance}}(j) + b_{\text{crowd}}(j)\}, \qquad P(i \to j) = \frac{w_{i\to j}}{\sum_{k\in N(i)} w_{i\to k}}.$$

Here i is the current room, j a neighbor, $N(i)$ the set of legal next rooms and $P(i \to j)$ the one-minute movement probability. The three b terms are local pulls: route following, room value and

Table 1: Museum-wide simulator results: status quo versus the all-11 intervention world. Welfare is total completed visitor welfare. Revenue is daily ticket revenue in EUR.

Crowd	Welfare base	Welfare int.	Gain	Revenue base	Revenue int.
500	277,256	343,507	+24%	11,542	11,317
2,500	1,106,175	1,274,045	+15%	56,604	56,544
5,000	1,514,458	1,986,425	+31%	96,502	94,792

crowd avoidance. The 0.01 floor keeps a legal doorway from becoming impossible. The executable simulator adds policy-specific logic for RAMA slots, exit seeking, tour groups and Caravaggio priority. The model is behavioral: it reproduces plausible crowd pressure, while the RL environments solve the individual optimization separately.

3.2 Museum interventions

The intervention catalog contains access controls, price instruments, time architecture and demand redistribution. The final scenario is called "all 11" because the publication catalog has eleven intervention specifications, including two pre-bundled combinations. In code these are merged with last-write-wins semantics into nine distinct active switches: RAMA, secondary-attractor enrichment, extended hours, dynamic pricing, resident annual-pass expansion, timed entry, quiet hours, a tour-group cap of 10 and a EUR150 group surcharge. A separate portfolio optimizer screens interventions individually and then runs greedy forward selection with a one-step local search on a scalarized welfare/revenue objective (Appendix E). That optimization is an economic sensitivity check. The RL booking agents are trained in the all-11 world because it is the environment in which RAMA access is active and internally consistent.

Figure 1 gives the corresponding spatial evidence. At the 5,000-visitor crowd, Botticelli peak density falls from 1.82 to 1.00 and traffic is distributed over a longer operating window. The intervention does not empty the museum but keeps the highest-demand rooms closer to usable capacity.

4 MDP Formulations

Each individual agent is a discounted Markov decision process $\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, r, \gamma)$. For a visitor profile p, the agent maximizes

$$J(\pi) = \mathbb{E}_{\pi, P_p}\left[\sum_{t=0}^{T} \gamma^t \, r_p(S_t, A_t)\right],$$

where π is the visitor policy, P_p the simulator transition rule for profile p, T the visit horizon, γ the discount and r_p the profile-specific reward. The objective scores the whole visit, so delayed events such as a missed slot or a late exit still count. The full simulator state is Markovian only when it includes room, clock, crowd tensor, reservations, visit history, appreciation progress and exit slack. Each agent sees a compact observation $o_t = \phi(S_t)$, so the implemented agents are engineered approximations to a partially observed visitor problem. The relevant state and action space change with the decision problem. Choosing what the agent could observe and which action it was allowed to take mattered at least as much as choosing PPO or DQN.

4.1 Tabular proof of concept

The toy museum has 12 rooms and deterministic crowd waves. It is small enough for a literal action-value table. The state is

$$s = (\text{room}, t_{\text{bin}}, c_{\text{Bott}}, c_{\text{Leo}}, m_{\text{visited}}),$$

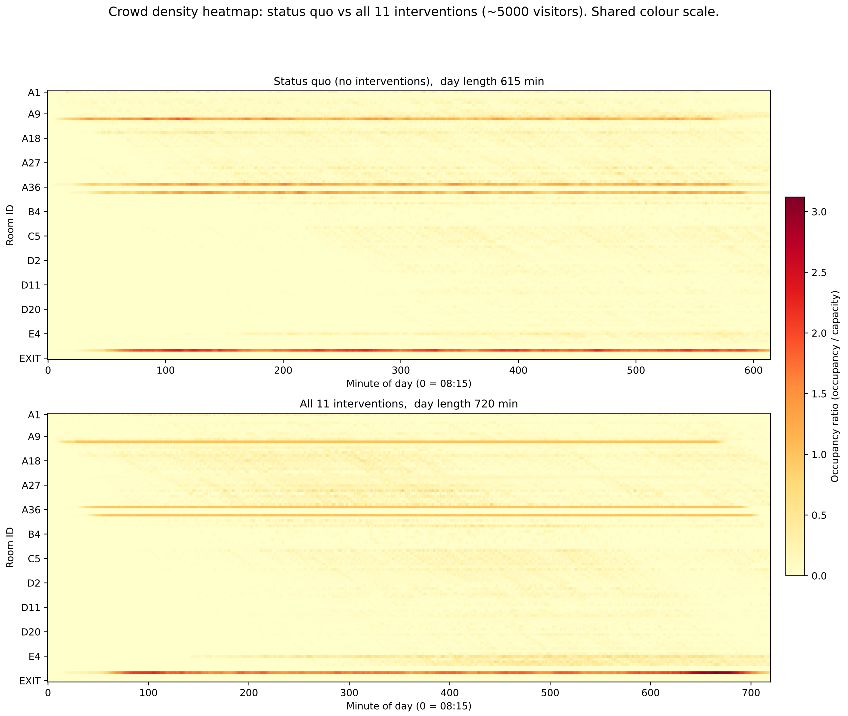

Figure 1: Crowd density under the status quo and under the full intervention bundle. The color scale is shared, so cooling of the red bands corresponds to a real reduction in occupancy over capacity.

where c_{Bott} and c_{Leo} are four-level crowd bins and m_{visited} is a bitmask over must-see rooms. That bitmask lets the table tell a half-finished visit from a completed one even when the visitor stands in the same room at the same time. Q-learning applies the masked Bellman update

$$Q(s,a) \leftarrow Q(s,a) + \alpha\Big[r + \gamma \max_{a' \in \mathcal{A}(s')} Q(s',a') - Q(s,a)\Big],$$

where $\mathcal{A}(s')$ contains only legal graph moves and the bracket is the temporal-difference error. Its per-step reward scores importance discounted by crowd density, with satiation and a small step cost. The full form is in Appendix B. The toy confirms the basic economic idea: learning beats fixed rules because it discovers temporal arbitrage, visiting Botticelli off-peak rather than mechanically following the route. Over the five reported training seeds, Q-learning reaches 83.3 ± 3.8 mean return and Double-Q reaches 83.1 ± 3.7, while the handcrafted rules range from -72.0 to 35.4. The small Q versus Double-Q gap is expected: the toy is mostly deterministic, so there is little noisy maximization bias for Double-Q to remove (the bias and the Double-Q update are discussed in Appendix A).

4.2 Status-quo navigation

The status-quo full-museum agent uses a planned-route wrapper. It does not pick arbitrary neighbors. Instead it decides whether to remain in the current target or continue to the next room in the planned walk. This gives the booking agent a fair comparison: the baseline is optimized but cannot use RAMA. The observation vector is

$$o_{\text{walk}} \in \mathbb{R}^8 = (\text{drain}, \text{importance}, \rho, \text{on-target}, \text{plan-share}, t, \text{egress}, \text{distance-to-next}),$$

that is, normalized time pressure, room value, density, whether the agent is on its planned target, route progress, clock time, exit slack and distance to the next target. The "drain" feature is the crucial one: appreciation already extracted from the current room. Without it, the decision "stay until I have seen enough, then leave" is partially observed: the same Botticelli room and clock time can require opposite actions depending on how long the visitor has already been absorbing the work.

Table 2: The three MDP families used in the project.

Setting	State	Action	Algorithms
Toy museum	Discrete 5-tuple: room, time bin, Botticelli crowd bin, Leonardo crowd bin, visited mask	Stay or move to a neighbor	Q-learning, Double-Q
Status-quo walk	8 real features: drain, importance, density, on-target, plan share, time, egress, distance	Stay or move on	PPO, DQN
RAMA booking	25 real features: booking phase, lead screen, time of day, held windows, check-ins, access read, arrival priors, room tail	Phase-dependent lead, commit, stay, move choices	MaskablePPO, PPO, DQN

4.3 RAMA as a two-phase MDP

The RAMA environment is the centerpiece. It wraps the same core museum reward but changes the action interface. Phase 1 is a booking screen: choose a lead time from

$$\ell \in \{1, 7, 21, 35\} \text{ days},$$

then commit to masterpiece windows. Phase 2 is the walk, where the visitor stays or moves and must arrive inside the booked windows. The 25-dimensional observation has seven groups: booking position (3, the phase flag, booking step and lead chosen), slots open by lead (4), time of day (1), held windows (3) and check-ins (3), the access read of the current room (4), the frozen expected arrival times (3) and the local room tail (4, drain, value, density and egress). A check-in is recorded when the visitor reaches a booked room during its reserved ten-minute slot, so 3/3 means all three booked masterpieces were reached on time.

The simplified availability rule is crowd-dependent:

$$\text{available}(g, \ell) = \mathbb{1}\left[\ell \geq L_{\max}\, p_g\, \frac{N_{\text{day}}}{2500}\right],$$

where g indexes a masterpiece group, ℓ is the selected lead time, p_g is the group's popularity, N_{day} is daily demand and $L_{\max}$ sets the hardest lead-time threshold. As N_{day} rises the same room requires earlier booking, which is the mechanism that makes late booking fail on busy days. Botticelli, Leonardo and Raphael/Michelangelo use ten-minute slot ledgers in rooms A11, A35 and A38. Caravaggio stays free-flow. The policy is never told "book early when busy." It sees only whether slots are still open and later receives the visit reward.

A scoring mistake surfaced at exactly this point. An early welfare metric charged a visitor for any masterpiece slot that had already sold out by the time it tried to book. That made the target unreachable on busy days and punished the agent for the state of the museum rather than for its own choices. We changed welfare to score the visit that was still attainable at the moment of booking, which is also what makes the gains in Section 6 comparable across crowd levels.

5 Reward Engineering

The deep agents share a core per-minute reward implemented in the full Uffizi environment:

$$r_t = r_{\text{art}} + r_{\text{time}} + r_{\text{close}} + r_{\text{complete}} + r_{\text{bored}} + r_{\text{checkin}} + r_{\text{wait}}.$$

The terms score art appreciation, a small time cost, pressure to exit before closing, completion, a post-satiation boredom term, RAMA check-ins and waiting. The main term is art value with crowd discount and satiation:

$$r_{\text{art}} = I_i\, \frac{1}{1 + \alpha \rho_i^2}\, \text{clip}\left(\frac{D_i - e_i}{s}, 0, 1\right), \qquad D_i = \max\{1, k_d(I_i - I_0)\}.$$

Here I_i is the visitor-specific room importance, ρ_i is density over capacity, e_i is appreciation already extracted, α controls crowd aversion, s scales satiation and D_i is the ideal dwell time. Appreciation accumulates more slowly in crowded rooms, so congestion hurts twice: it lowers instantaneous utility and consumes more of the fixed day. A famous room still pays off when crowded, but the payoff vanishes once the visitor has extracted the room's value, so the formula discourages overcrowding without teaching the agent to skip Botticelli.

The final reward is the product of several failed intermediate specifications. The failures are the most instructive part of the project. A purely terminal penalty for not exiting produced an almost flat reward surface. The agent often learned to sit in one room until closing. The fix was a dense closing signal,

$$r_{\text{close}} = -\kappa \max(0, d_{\text{exit}} + b - \tau),$$

where d_{exit} is walking time to an exit, b a safety buffer, τ the time left and κ the penalty slope. The term is zero while the visitor has slack and turns negative only when the remaining time is too short for a safe exit, which gives gradient descent a slope to follow toward the door before closing.

Crowding had to be handled conservatively. When congestion was penalized too harshly, the art lover learned a degenerate policy that avoided crowded masterpieces altogether: a crowded Botticelli scored worse than skipping it. The final full-museum reward, therefore, has no separate explicit congestion penalty. Density discounts art value smoothly without dominating it, because a crowded Botticelli should still beat an empty corridor. Strong boredom penalties were just as bad, because moving reset the local penalty and the agent learned to loop. The final reward relies mainly on satiation and a small step cost, with only a modest post-satiation boredom term active in the booking wrappers.

RAMA adds reservation-specific penalties, the first two charged during the booking phase and the third once at the end of the episode:

$$r_{\text{book}} = -d_{\text{decline}} n_{\text{decline}} - k_{\text{mm}} \sum_{g \in \mathcal{B}} |w_g - \hat{t}_g| - d_{\text{noshow}} n_{\text{noshow}}.$$

Here $\mathcal{B}$ is the booked set. The term n_{decline} counts skipped reservations, charged the moment a masterpiece is declined. The mismatch term penalizes the gap between each reserved window w_g and the frozen expected arrival $\hat{t}_g$, charged at commit. The term n_{noshow} counts masterpieces booked but never reached in their window, charged once when the episode ends. The d, k constants set the penalty sizes. The mismatch term is what made booking learnable: it charges a bad 11:00 Botticelli slot immediately if the reference walk reaches A11 near 10:20, instead of letting the cost of a bad reservation arrive hundreds of steps later. We also encode "books the three masterpieces" as a visitor-type trait by removing the free-decline option (`allow_decline=False`). Otherwise a decline action creates a degenerate solution in which the agent dodges delayed risk by reserving nothing at all.

6 Experiments and results

6.1 Booking policy

Table 3 shows the individual RL result used in the figures. The baseline is the best matched no-intervention walk. The intervened score is the MaskablePPO RAMA booking policy in the all-11 world, averaged over five independent training seeds and six fixed evaluation seeds. The art lover at the packed crowd is the one visibly unstable cell: four runs learn the intended 3/3 booking pattern while one collapses, which produces both the large standard deviation and the 2.6/3 mean check-in rate. We report the five-reseed mean precisely because it keeps that failure visible. A single favorable model reaches 6,297 points at 2,500 visitors and 5,646 at 5,000 (Appendix F).

The tourist result is more stable: a shorter checklist and shorter dwell targets make it easier to book the three windows and execute the route. The learned lead-time curve has the intended comparative static. Art lovers move from 7-day booking at the quiet crowd to 35-day booking at

Table 3: RAMA booking agent versus matched no-intervention walk baseline. Points are five-training-seed means. Parentheses report the standard deviation of the intervened arm.

Profile	Crowd	Baseline	RAMA	Gain	Check-ins
Art lover	500	5,112	7,113 (0)	+39%	3.0/3
Art lover	2,500	4,780	6,167 (291)	+29%	3.0/3
Art lover	5,000	4,010	4,800 (1,891)	+20%	2.6/3
Tourist	500	2,694	3,686 (0)	+37%	3.0/3
Tourist	2,500	2,708	3,695 (0)	+36%	3.0/3
Tourist	5,000	2,680	3,685 (0)	+38%	3.0/3

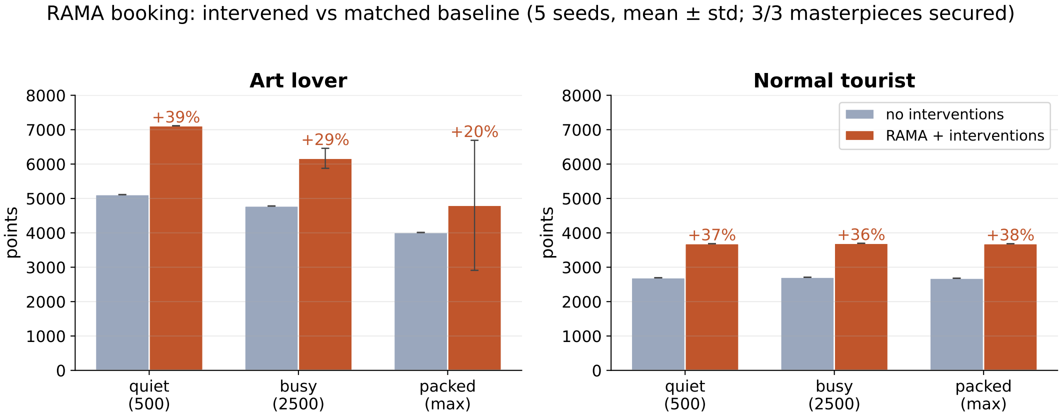

Figure 2: RAMA booking gains over matched no-intervention walk baselines, reported as five-training-seed means with standard-deviation bands. The grid shows both profile heterogeneity and crowd-level dependence.

the busy and packed crowds. Tourists stay at 7 days until the packed crowd, where the reseeded curve moves to 21 days. None of this is hard-coded: the agent sees only slot availability and the downstream reward.

6.2 Algorithm comparison

Figure 3 isolates the algorithmic conclusion. MaskablePPO is best or tied in every cell. Unmasked PPO collapses for the packed tourist in the intervened world, falling to 2,093 points where MaskablePPO reaches 3,685. DQN is competitive in several easy tourist cells but crowd-fragile for the art lover. Under the no-intervention baseline its art-lover return falls from 4,527 at the quiet crowd to 2,044 at the packed crowd in the five-reseed mean. The pattern matches the algorithms' known weaknesses. DQN inherits a maximization step over noisy value estimates, the same issue that motivates Double-Q in the tabular setting. PPO avoids that max-over-actions target, but unmasked PPO still spends exploration budget discovering actions that are impossible in the current graph or booking phase. MaskablePPO removes that burden by matching the policy support to the legal action set at every minute. The masking machinery and the full settings are in Appendices A and C.

One evaluation choice mattered more than we expected. Scoring a stochastic PPO navigation policy by its greedy argmax action misread it, because the policy's value comes from its spread over routes rather than from a single deterministic path. We, therefore, evaluate booking deterministically, where the decision really is a single best action, while navigation diagnostics are sampled. To keep the comparison fair across policies we score on common random numbers. The protocol is in Appendix D.

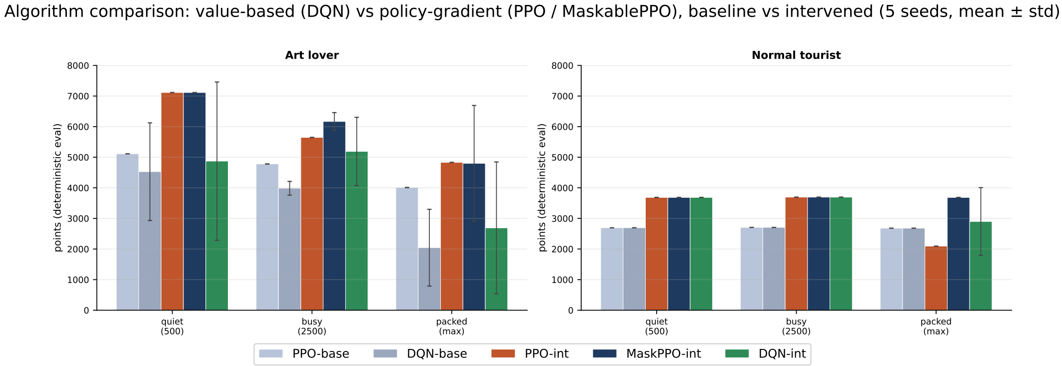

Figure 3: Algorithm comparison across profile and crowd cells. The dark-blue bars are MaskablePPO. The gap to unmasked PPO at the packed tourist cell is the cleanest empirical evidence for action masking.

7 Conclusion

We asked whether reinforcement learning can ease congestion at the Uffizi, both by redesigning access at the level of the institution and by guiding a single visitor through the crowd. The answer is yes on both counts. The harder half of the work was stating the problem correctly rather than choosing the algorithm.

At the museum level, a bundle of eleven Pigovian interventions raises completed welfare by 31% at the busiest crowd while holding revenue, cooling the worst density bands. At the visitor level, once RAMA turns the visit into a booking problem, a MaskablePPO agent beats its matched no-booking baseline in every profile and crowd cell and reserves earlier as the museum fills, a behaviour we never coded. Masking is decisive at the hardest cells, where unmasked PPO collapses and value-based DQN stays crowd-fragile.

Two implications follow. For the museum, the gains come from pricing the congestion externality rather than from adding capacity, so the levers are opening hours, access windows and crowd-sensitive prices a gallery can actually pull. For reinforcement learning, the leverage sat in the formulation: an observation carrying appreciation progress, a reward dense near closing yet gentle on crowds and a mask that encodes what is physically and contractually legal.

The work has clear limits. Topology and hours come from public sources and the official floor plan, but the 60/30/10 profile mix, the heterogeneity scale, the willingness-to-pay elasticity, the compliance rates and the RAMA fill curve are assumptions, so the numbers describe a controlled simulation rather than a forecast of the real Uffizi. The reported RAMA gain also mixes RAMA with the rest of the bundle, since its control is a no-intervention walk rather than an all-11 world without booking. "Always books the three masterpieces" is a type assumption rather than a learned preference. The agent faces an exogenous crowd rather than the full equilibrium, a mean-field problem we did not build. Welfare is a scalar proxy that leaves fairness and operations aside.

Overtourism is usually fought with blunt caps and turnstiles. Our results point to a finer instrument: the redesign shifts visitors in time rather than turning them away, while a booking policy gets more out of a crowded day than a fixed route allows. The plague of the famous room is that everyone wants it at once. Pricing that congestion and letting visitors book around it recovers most of what the crowd destroys. The agent that learns to do so is only as good as the problem we wrote down for it.

A Algorithms and masking

The full museum is too large for a table, so we use PPO-family actor-critic methods for the main agents and DQN as the value-based control.

PPO. PPO [4] optimizes the clipped surrogate

$$L^{\mathrm{CLIP}}(\theta) = \mathbb{E}_t\left[\min\left(\rho_t \hat{A}_t,\, \mathrm{clip}(\rho_t, 1-\epsilon, 1+\epsilon)\hat{A}_t\right)\right], \qquad \rho_t = \frac{\pi_\theta(a_t \mid s_t)}{\pi_{\theta_{\mathrm{old}}}(a_t \mid s_t)}.$$

Here θ is the policy network, ρ_t compares the new and old policy on the action actually taken, $\hat{A}_t$ estimates whether that action did better than expected and ϵ limits how far one update can move the policy. The clip protects the policy from overreacting to a single lucky route through a quiet Botticelli room.

Generalized advantage estimation. The advantages use GAE,

$$\hat{A}_t = \sum_{l \geq 0} (\gamma\lambda)^l \delta_{t+l}, \qquad \delta_t = r_t + \gamma V_\phi(s_{t+1}) - V_\phi(s_t),$$

where V_ϕ is the learned value baseline and λ controls how much the estimate trusts long rollouts ($\lambda \to 1$) versus the critic's one-step guess ($\lambda \to 0$). GAE smooths the noisy minute-by-minute rewards over the visit while still reacting to a missed slot or a late exit.

DQN. DQN [3] minimizes a squared Bellman error against a target network and a replay buffer:

$$L(\theta) = \mathbb{E}_{(s,a,r,s') \sim \mathcal{D}}\left[\left(r + \gamma \max_{a'} Q_{\theta^-}(s', a') - Q_\theta(s, a)\right)^2\right].$$

The replay buffer $\mathcal{D}$ is a rolling memory of experience tuples (s, a, r, s'). Training on random minibatches from it reduces correlation between updates. θ is the current value network and θ^- a slower target network. The $\max_{a'}$ over noisy estimates is the source of the overestimation that makes DQN crowd-fragile in our results.

Double-Q. Double-Q [5] addresses that overestimation by separating action selection from action evaluation:

$$Q_A(s, a) \leftarrow Q_A(s, a) + \alpha\Big[r + \gamma Q_B(s', \arg\max_{a' \in \mathcal{A}(s')} Q_A(s', a')) - Q_A(s, a)\Big],$$

with the symmetric update for Q_B. The A table chooses the next room through the $\arg\max$ and the B table evaluates that choice, which avoids repeatedly overpricing the same tempting room. In the deterministic toy the two methods are almost tied, as Section 4 reports.

Action masking. In the full navigation environment most graph-action indices are illegal in most states. In the RAMA wrapper the action space is smaller but phase-dependent, since lead choices are legal only on the booking screen, commit choices only at the reservation step and stay/move choices only during the walk. MaskablePPO sets illegal logits to $-\infty$ before the softmax,

$$\pi_\theta(a \mid s) = \frac{\exp(z_a)\mathbb{1}[a \in \mathcal{A}(s)]}{\sum_{b \in \mathcal{A}(s)} \exp(z_b)},$$

where z_a is the raw network score for action a and $\mathcal{A}(s)$ is the current legal set. Invalid actions receive exactly zero probability and zero gradient, so a visitor in A11 cannot assign probability to a nonexistent door or to a booking-screen action during the walk. Unmasked PPO must instead learn legality by sampling illegal moves and paying penalties, which wastes data. In easy cells the difference is small, in the packed tourist RAMA cell it is outcome-determining.

B Toy reward

The 12-room toy uses a per-step Type-A reward

$$r^{\text{toy}} = \frac{I\,\nu}{1+\alpha\,\rho^2} - c_{\text{step}},$$

where I is room importance, ν is novelty that halves on each extra minute spent in the same room (so the room satiates), ρ is the synthetic crowd density, $\alpha = 6$ and $c_{\text{step}} = 0.5$. Three flat terms complete it: a congestion penalty $-0.5\,\rho$ when $\rho > 0.8$ in a gallery, an invalid-action penalty -0.2 and, at the end, an exit bonus $+15$ for a voluntary exit against a no-exit penalty -50 if the visitor is still inside at the horizon. Only the relative sizes matter. The values are hand-tuned.

C Training hyperparameters

The PPO and MaskablePPO agents use Stable-Baselines3 MLP policies with two 128-unit hidden layers. They collect 2048 museum steps per rollout, split them into batches of 256 and pass over those batches 10 times per update. The learning rate is 3×10^{-4}, GAE uses $\lambda = 0.98$, the clip range is 0.2 and the entropy coefficient is 0.01. The no-booking walk baselines train for 150k steps with $\gamma = 0.995$ for art lovers and 0.997 for tourists. RAMA booking policies train for 100k steps with $\gamma = 0.999$, which keeps the delayed reservation payoffs alive. DQN uses a 100k replay buffer, a target update every 1000 steps, train frequency 4, a 2k-step warmup before learning starts and ε-greedy exploration decaying from 0.30 to 0.05.

Table 4: Hyperparameters by arm. One simulator step is one minute, so a full visit is the 720-minute day plus four booking decisions, about 720 steps, so 100k steps is roughly 140 visits and a 2048-step rollout is about three.

Arm	Net	Learning rate	γ	Budget
Toy Q / Double-Q	tabular	$\alpha = 0.1$	0.99	25k episodes
PPO baseline (walk)	$[128, 128]$	3×10^{-4}	0.995/0.997	150k steps
MaskablePPO (RAMA booking)	$[128, 128]$	3×10^{-4}	0.999	100k steps
PPO intervened (no mask)	$[128, 128]$	3×10^{-4}	0.999	100k steps
DQN (baseline / intervened)	$[128, 128]$	5×10^{-4}	0.995–0.997 / 0.999	150k / 100k

For readers new to the knobs: n_{steps} is the experience collected before each learning phase, the batch size is the chunk used per weight step and an epoch is one full pass over the collected batch. γ sets how far ahead reward counts. The value 0.999 keeps a payoff hundreds of steps away alive for booking. GAE λ trades off trusting long rollouts against the critic's one-step estimate. The clip range is PPO's brake on how fast a single update can change an action's probability. The entropy coefficient is a small bonus for keeping the policy spread out, so it explores before committing. The toy uses ε decaying from 1.0 toward a small floor by $\times 0.9995$ per episode.

D Evaluation protocol

Evaluation uses common random numbers. A training seed changes network initialization and exploration. A crowd seed fixes one simulated visitor day. We train five policies with seeds $\{1, 3, 7, 42, 202606\}$ and score each on the same six crowd seeds, `900000`–`900005`, using the highest-probability legal action rather than sampling. The tables first average the six crowd days for each trained policy, then report the mean and spread across the five policies, which keeps instability visible instead of hiding it behind one favorable run. Booking is scored deterministically because the booking decision is genuinely a single best action. Navigation diagnostics are sampled, because a greedy argmax misreads a stochastic navigation policy whose value comes from its spread over routes.

E Institutional screening and portfolio optimization

The director gate is the constrained screening rule

$$\mathcal{I}^* \in \arg\max_{\mathcal{I}\in\Omega} W(\mathcal{I}) \quad \text{s.t.} \quad R(\mathcal{I}) \geq (1-\delta)R(0),$$

where $\mathcal{I}$ is an intervention bundle, Ω the set of bundles we test, W simulated completed welfare, R ticket revenue, $R(0)$ status-quo revenue and δ the allowed revenue loss. The constraint rejects an intervention that improves the visit only by giving up too much revenue. The portfolio optimizer first screens the eleven catalog interventions individually, then runs greedy forward selection followed by a one-step local search on a scalarized welfare/revenue objective. This is an economic sensitivity check on the bundle. The RL booking agents are trained in the all-11 world because that is where RAMA access is active and internally consistent.

F Gain definition and the deliverable run

The gain column in Table 3 is

$$\text{Gain} = 100 \times \frac{J_R - J_B}{J_B},$$

where J_R is the RAMA mean return and J_B the matched walk return without RAMA, so a +39% gain means the RAMA policy earns 39% more than its matched baseline. The parentheses are standard deviations across RAMA training seeds. The baseline is a fixed comparator in each profile-crowd cell, so its own reseed spread is not reported. The selected single-model deliverable run is stronger in the art-lover high-crowd cells, reaching 6,297 points at 2,500 visitors and 5,646 at 5,000 in the six-seed deterministic evaluation. The five-reseed table in the main text is lower and more informative, because it exposes the one art-lover run at the packed crowd that collapses while the other four learn the 3/3 pattern.

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